

Hovercraft Hockey Hackathon

Build a puck that moves by barely touching the floor.

Win condition: Highest normalized glide and five-shot target scores, with explanation and build quality.

THE SCIENCE YOU ARE USING

Air cushion cuts friction. A balloon pushes air down through a hole in a disc, lifting the disc on a thin cushion of air. With almost no contact, friction nearly vanishes, so a small push sends it gliding. Newton's laws then keep it moving until something stops it.

Glide versus control. Maximum glide is easy; control is hard. You balance lift and steerability to compete in gym-safe target hockey. Build quality and a clear explanation also score.

HOW TO BUILD AND RUN IT

- 01 Mount the valve.** Ask a staff member to use low-temperature hot glue to attach a pop-top cap over the center hole of the disc and seal the edge.
- 02 Attach the balloon.** Stretch an inflated balloon over the closed cap so air escapes only through the hole.
- 03 Test the glide.** Open the cap just a little, set it down, and give a gentle push. A small opening glides longer and steers better; open wider only if it will not lift.
- 04 Practice control.** Learn to start and stop on target. Too much lift means no control.
- 05 Play target hockey.** Use the Glide Test and Target Rules sheet. Glide points are normalized to the field-best distance out of 30; divide the five-shot target total by 6 and round for a maximum of 25.

Safety: Low-temperature hot glue is staff-only; wait until it cools before handling the puck. **Latex allergy note.** No face-level launches. Play floor-only. A grown-up inflates the balloons; throw away any balloon that pops right away. Allergy: Latex balloon alternative: small bags or school-approved nonlatex balloons.

MATERIALS

Recycled CDs or discs	per team
Pop-top bottle caps	per team
Balloons	per team
Low-temperature hot glue	staff station
Targets	shared
Floor tape	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Glide performance	30
Target competition score	25
Explanation	20
Redesign	15
Build quality	10
Total	100